

John Francis Burkhart

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Education

Northern Arizona University

B.S. Mathematics, Summa Sum Laude. GPA: 4.0/4.0

Flagstaff, AZ

Aug. 2018 - May 2022

Research Experience

North Carolina State University

Raleigh, NC

Graduate Research Assistant

Aug. 2023 - April 2024

- Co-authored and submitted paper (under review) on optimizing genome sampling for demographic and epidemiological inference with Markov Decision processes (MDPs).
- Formulated MPD to optimize genome sampling for estimation of growth rate parameter in an exponentially growing population under coalescent model.
- Used Python, R, and bash scripts to prepare genetic data for analysis of pathogen phylogeny.
- Prepared medical image data for computer vision tasks (segmentation and classification).
- Completed two semesters of graduate coursework in genetics, statistics, and computer science.

Northern Arizona University

Flagstaff, AZ

Research Technician

Aug. 2022 - Jul. 2023

- Developed NIH-funded, Python-based [open source epidemiological modeling software](#).
- Followed software engineering best practices by utilizing object-oriented design patterns, Git for version control, GitHub actions for CI/CD, and well-written documentation for project continuity.
- Mentored math and computer science undergraduate researchers.

Undergraduate Math Assistant

Aug. 2021 - May 2022

- Studied the enumeration of signed permutations under the action of reversals and their applications to genome inversions.
- Solved several open problems, including finding a closed form solution for the number of signed permutations by reversal distance.
- Awarded \$5,000 Hooper Undergraduate Research Award.
- Presented results at three different conferences.

Machine Learning Lab Research Assistant

Aug. 2021 - May 2022

- Cleaned and implemented machine learning algorithms (Poisson regression, extreme gradient boosting, and random forests) on a large data set of multiplex polymerase chain reaction (mPCR) trials.
- Presented findings at undergraduate research symposium and published work to GitHub.

CUNY Baruch

Manhattan, NY (remote)

Baruch Discrete Math REU Participant

Summer 2021

- Adapted new methods for improving lower bound on Ramsey numbers to the multicolored bipartite setting.

Industry Experience

The CDC Foundation

Atlanta, GA (Remote)

Data Analyst

Sep. 2024 - Present

- Uses SQL, Python, and Tableau to extract, transform, load, and visualize non-profit fundraising data.
- Implements machine learning algorithms using Python (Pandas, Numpy, and Scikit-Learn) to aid in portfolio analysis and revenue forecasting.
- Automates time-consuming data quality and maintenance tasks using Python and Salesforce REST API.

ASM America

Intern Data Analyst

Phoenix, AZ (Remote)

Summer 2022

- Designed and built Power BI dashboards to visualize Global Product Team sales data

Teaching Experience

Northern Arizona University

Flagstaff, AZ

Calculus I Peer Teaching Assistant

Aug. 2020 - Dec. 2020

Linear Algebra Peer Teaching Assistant

Jan. 2020 - May 2020

Peer Math Assistant (Calculus I and III tutor)

Jan. 2019 - May 2020

Arizona Aspire Academy

Phoenix, AZ

Math and Music Tutor

Aug. 2017 - May 2018

Academic Writing

1. Optimizing genomic sampling for demographic and epidemiological inference with Markov decision processes. D. Rasmussen, M. Bursell, **F. Burkhardt**. Currently under review. [Preprint: *bioRxiv*, 2025.06.30.662264](#).
2. Enumerating signed permutations by reversal distance. F. Awik, **F. Burkhardt**, H. Denoncourt, D.C. Ernst, T. Rosenberg, and A. Stewart (in preparation).

Presentations

2023

1. The Enumerative Combinatorics of Genome Inversions for Bioinformaticians. Invited talk given to the Phylodynamics Research Group at NC State.

2022

1. The Enumeration of signed permutations Under the Actions of Reversals. Virtual poster presentation at the Joint Mathematics Meeting.
2. The Enumeration of signed permutations Under the Actions of Reversals. Poster presentation at the Northern Arizona University Undergraduate Research Symposium.
3. The Enumeration of signed permutations Under the Actions of Reversals. Poster presentation at the Hooper Undergraduate Research Award Symposium.

2021

1. Improving the Bounds on Multicolored Bipartite Ramsey Number. Slide show presentation at the CUNY Discrete Math REU.
2. Using mlr3 for Machine Learning Research. Invited talk at the NAU Machine Learning Lab.

2020

1. Machine Learning Algorithms for Multiplex Polymerase Chain Reaction Primer Design. Virtual Poster presentation at the Northern Arizona University Undergraduate Research Symposium.

Awards and Honors

- NIEHS Environmental Health Bioinformatics (EHB) Fellowship Aug. 2023
- NAU Department of Mathematics and Statistics Outstanding Senior May 2022
- J. Harvey Butchart Mathematics Scholarship Jan. 2021
- Hooper Undergraduate Research Award (HURA) Jan. 2021

Technical Skills

Favorite Languages and Tools: Python, R, SQL, Tableau, Git, and LaTeX.